BST Physics Curriculum Vol 6 Chapter 12 — Information Theory + Thermodynamics v0.4 (refilled per Cal #104)

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Vol 6 Chapter 12 — Information Theory + Thermodynamics

Chapter motivation

Standard Shannon-Landauer bridging connects information theory + thermodynamics: - Shannon entropy H(X) = −Σ p_i log p_i (information theory, Shannon 1948) - Boltzmann entropy S = k_B ln(W) = k_B · H(X) · ln(2) (thermodynamics, with Boltzmann-to-bits unit conversion) - Landauer's principle (1961): erasing one bit of information generates AT MINIMUM k_B T ln(2) of heat - Bérut 2012 experimental verification: Landauer bound confirmed in single-electron experiment - Maxwell's demon resolution: demon must record information; demon's information record has Shannon entropy; demon-system total entropy non-decreasing

BST per Casey CSE directive (Wednesday) identifies substrate as information substrate: substrate-tick GF(128)^k state IS information; entropy IS Shannon entropy of substrate states; Landauer cost = substrate-tick computational cost per Koons tick. Quantum-information bridging via K67 Born = Bergman RATIFIED + T2479 POVM Saturday.

Section 12.0b — Reader-grade 3-level pedagogy

Level 1: Shannon entropy H(X) ↔ thermodynamic entropy S (Vol 6 Ch 3); Landauer's principle (erasing 1 bit costs ≥ k_B T ln(2)) derives from substrate-tick computational cost per Koons tick; substrate IS information substrate per Casey CSE directive (Wednesday).

Level 2 (graduate-physicist): Shannon entropy H(X) = −Σ p_i log_2 p_i (in bits) ↔ Boltzmann S = k_B ln(W) (in Joules/Kelvin); bridge: S = k_B · H(X) · ln(2). Landauer's principle 1961: irreversible computation (erasing 1 bit) generates ≥ k_B T ln(2) heat — operationally confirmed Bérut et al. 2012 (Nature) using single-electron logic gates. BST substrate-cartography per Casey CSE directive (Wednesday): substrate IS information substrate; substrate-tick GF(128)^k state distribution IS Shannon-counting at substrate-tick scale; Landauer cost = substrate-tick computational cost per Koons tick (~10^{-120} s, T2405). Maxwell's demon resolution: demon must record information (substrate-tick GF(128)^k state); demon's information record has Shannon entropy; demon-system total entropy non-decreasing per substrate-Shannon-counting. Quantum information bridging via T2479 POVM (Saturday SP-31 #283 closure) + K67 Born = Bergman: quantum entropy S_vN = $-\mathrm{Tr}(\rho \ln \rho)$ substrate-cartography reading. Vol 14 Information Theory full cross-link: Vol 14 Ch 3 Shannon channel capacity + Ch 5 Born=Bergman + Ch 12 substrate-CI architecture INTERNAL register. Reversible-computation substrate-cartography: cyclotomic-projection-respecting operations preserve information (K59 RATIFIED 7-step chain); cyclotomic-violating operations cost ≥ k_B T ln(2) per bit per Landauer.

Level 3 (5th-grader accessible): Information theory and thermodynamics are linked: Shannon entropy (information bits) and thermodynamic entropy (heat/temperature) are the same concept in different units (k_B · ln(2) conversion). Landauer's principle says erasing one bit of information generates at least k_B T ln(2) of heat (proven experimentally in 2012 by Bérut et al. using single-electron experiments). BST per Casey CSE directive (Wednesday) identifies the substrate as an information substrate — substrate-tick GF(128)^k states ARE information; thermodynamic entropy IS Shannon entropy at substrate scale. Maxwell's demon is resolved: the demon must record information, the demon's records have entropy, total entropy is conserved.

Section 12.1 — Shannon-Boltzmann Bridge

Shannon 1948: H(X) = −Σ p_i log_2 p_i (information theory, units of bits). Boltzmann: S = k_B ln(W) (thermodynamics, units of J/K). Bridge: S = k_B · H(X) · ln(2) — unit conversion.

Boltzmann formula S = k_B ln(W) where W = number of microstates consistent with macroscopic constraints IS Shannon entropy at substrate scale with W = 2^{H(X)} per binary substrate-tick framing.

Section 12.2 — Landauer's Principle (1961)

Landauer 1961: irreversible computation (1-bit erasure) generates AT MINIMUM k_B T ln(2) ≈ 2.85×10^{-21} J at room temperature. Sets fundamental lower bound on energy cost of computation.

Bérut et al. 2012 (Nature): experimental verification using single-electron Brownian particle in double-well potential; measured heat dissipation at Landauer bound.

Section 12.3 — BST Substrate as Information Substrate (Casey CSE Wednesday)

Casey CSE directive Wednesday: substrate IS information substrate (Computational Science Engineering framing). Substrate-tick GF(128)^k state distribution IS Shannon-counting at substrate-tick scale.

Substrate-tick framing: - Per Koons tick (~10^{-120} s, T2405): substrate state ∈ GF(128)^k = 2^{7k} possible states - Per-tick Shannon entropy ≤ 7k bits (substrate-tick UV cutoff) - Landauer cost = substrate-tick computational cost = energy expended per Koons tick to compute next state

Section 12.4 — Maxwell's Demon Resolution

Maxwell's demon (1867 thought experiment): hypothetical entity opens/closes gate between two gas chambers based on molecule speed; appears to violate second law by sorting fast-cold molecules.

Resolution (Bennett 1973 + Landauer 1961): demon must RECORD information about each molecule; demon's information storage has Shannon entropy; demon-erasure cost = k_B T ln(2) per bit ≥ thermodynamic entropy gain from sorting.

BST substrate-cartography: demon's information record is substrate-tick GF(128)^k state; demon-erasure substrate-tick cost matches Landauer bound exactly per substrate-Shannon-counting.

Section 12.5 — Quantum Information Bridging (K67 + T2479)

K67 Born = Bergman RATIFIED Tuesday: quantum probability = Bergman reproducing-kernel evaluation; bridges quantum measurement to substrate information extraction.

T2479 POVM substrate-derivation Saturday: generalized quantum measurements substrate-cartography via Bergman positive-definiteness + T2417 4-Zone observer bandwidth.

von Neumann entropy S_vN = $-\mathrm{Tr}(\rho \ln \rho)$ substrate-cartography reading: substrate-tick K-type basis density-matrix Shannon entropy.

Section 12.6 — Reversible Computation Substrate-Cartography

Cyclotomic-projection-respecting operations (K59 RATIFIED 7-step chain) preserve information at substrate-tick level — reversible computation at substrate scale.

Cyclotomic-violating operations cost ≥ k_B T ln(2) per bit per Landauer bound — irreversible computation.

Quantum-computer reversibility: unitary gates preserve information; measurement is irreversible (Landauer-equivalent).

Section 12.7 — Honest scope + Vol 14 Cross-Link

Vol 14 Information Theory full cross-link: Vol 14 Ch 3 Shannon channel capacity + Ch 4 Nyquist sampling + Ch 5 Born=Bergman + Ch 9 Kolmogorov + Ch 12 substrate-CI architecture INTERNAL register.

Falsifier: Landauer bound violation in physical computation → BST refuted at thermodynamics-information bridge level.

Open scope (multi-week): - Quantitative substrate-tick Landauer-cost derivation from K59 7-step chain - Quantum-information substrate-mechanism completeness verification - Vol 8 Biology of Mind cross-link (substrate-coupled biological observers)

Section 12.8 — Vol 6 Status Summary

Vol 6 Thermodynamics & Statistical Mechanics 12/12 chapters at v0.4 chapter-grade narrative COMPLETE (Saturday 2026-05-23 EDT Wave 2 third volume; Cal #104 refill complete for Ch 6-12).

Cross-CI handoff: Grace Vol 6 12 catalog backbones; Elie cross-CI verification toys; Cal cold-read pipeline; Keeper K222-K233 K-audit pre-stages.

— Lyra, Vol 6 Ch 12 v0.4 chapter-grade narrative REFILLED per Cal #104; Vol 6 12/12 COMPLETE at v0.4 refilled, Saturday 2026-05-23 EDT